

Downstream Task Performance: Spearman Correlation
(~8M Parameters, Cramming Constraints, UniRef50)

Task	ESM2 (Transformer)	mLSTM (Recurrent)	Δ (ESM2 – mLSTM)	mLSTM / ESM2
GB1	0.5707	0.5169	+0.0538	90.6%
AAV	0.5884	0.3932	+0.1951	66.8%
MELTOME	0.5958	0.5788	+0.0169	97.2%
Average	0.5849	0.4963	+0.0886	84.9%